

Confidence-based Filtering for Enhancing Recommendation Quality

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Abstract. In this article, we present an algorithm that leverages rating prediction confidence factors to enhance recommendation quality. More specifically, this algorithm filters candidate recommendation items, by removing those whose rating predictions do not fulfil any rating prediction confidence factor and subsequently proceeds in formulating the recommendation(s) with the remaining candidate items. Under this approach, only rating predictions that have increased accuracy guarantees are utilized, and therefore the quality of the recommendation is enhanced. The presented approach extends previous works, and is proven to maintain high recommendation coverage. At the same time it can be applied to every collaborative filtering dataset, since it is based only on the essential collaborative filtering data (i.e., rating records which include the user id, the item id and the respective rating). To evaluate the proposed extension, datasets commonly used in the recommender systems research field are used. The evaluation results indicate its effectiveness towards recommendation quality, outreaching the performance of state-of-the-art collaborative filtering research works that also consider rating prediction confidence factors. The presented approach can finally be easily combined with collaborative filtering recommendation algorithms that also consider the rating prediction confidence factors.

Keywords: Recommender Systems, Collaborative Filtering, Recommendation Quality, Rating Prediction Confidence Factors, Multi-Parameter Experimental Evaluation.

1 Introduction

Collaborative Filtering (CF) is one of the most commonly used techniques for predicting rating values in recommender systems (RecSys). This method relies on the ratings given by users who are closely aligned with the active user (AU), referred to as

near neighbors (NNs). Typically, a CF algorithm first generates rating predictions for products the AU has not yet rated. The products with the highest predicted ratings are then recommended to the AU, as they are most likely to be well-received. Therefore, the more accurately these predicted ratings match the actual ratings, the more successful the CF RecSys is proven to be [1–4].

For instance, if two products p_1 and p_2 are candidate for recommendation to an AU u , with rating predictions calculated at $pr_1 = 4.95/5$ and $pr_2 = 4.85/5$, respectively, a typical CF RecSys would firstly select to suggest p_1 to u . This is because its higher predicted rating suggests a greater likelihood of acceptance by u .

However, in the previous example, the information used to calculate rating predictions is not fully considered. Specifically, while a typical CF RecSys computes additional data related to each rating prediction -such as the number of near neighbors (NNs) involved in the rating prediction for each product- this information is not examined during the recommendation process. For instance, if pr_1 is calculated using only 1 of the u 's NNs, while pr_2 is based on 25 of the u 's NNs, there is a valid reason to suggest product p_2 over p_1 . The rationale behind this option is that p_2 , besides having a very high prediction value, is also produced from a relatively large number of NNs. This entails higher probability of p_2 recommendation accuracy

Recent research has explored the concept of rating prediction confidence factors (RPCFs), particularly in relation to the accuracy of rating predictions derived from the CF process [5,6]. Specifically, these studies have demonstrated that (1) the number of NNs involved in the prediction calculation, (2) the mean rating score of the products, and (3) the mean rating score of the AU, are directly linked to the accuracy of rating predictions generated by the CF process.

These findings point to the direction that the CF RecSys can opt to recommend products with lower rating prediction values but higher prediction confidence and thus improve the overall recommendation quality. In this context, the work in [7] introduces a state-of-the-art algorithm that partitions the set of items in the list of items which are candidate for recommendation (termed as “initial recommendation candidate list”-ICRL) into three subsets depending on their CF prediction values. These three subsets are termed as “Top”, “Med” and “Low” and respectively comprise the rating predictions that are believed to be “definitely,” “most probably,” or “probably” liked by the user. The items within each of these subsets are then sorted in descending order based on the number of RPCFs satisfied, and the items are recommended to the AU, starting from the items in the first class, which are scanned according to their sort order, and then proceeding with the items in the next two classes, in the same fashion.

The present work introduces an extension to the algorithm presented in [7] that leverages RPCFs to further enhance recommendation quality. The proposed extension removes from the ICRL the items whose predictions do not fulfil any confidence factor, irrespectively of their rating prediction values, and subsequently proceeds in formulating the recommendation with the remaining elements of the ICRL. Under this approach, only predictions that have increased accuracy guarantees are utilized, and therefore the quality of the recommendation is enhanced. The proposed algorithm adopts a conservative approach to item removal from the ICRL, achieving the maintenance of high recommendation coverage. At the same time, it can be applied in every CF dataset, since

it is based only on the essential CF data (i.e., rating records which include the user id, the item id and the respective rating), and it can be combined with any rating prediction computation method.

A multi-parameter set of experiments is used to thoroughly assess the proposed extension for recommendation quality. This experiment includes (a) 8 CF datasets from various sources, (b) 2 user vicinity measures, (c) 2 NN selection alternatives, and (d) 3 recommendation quality metrics, all of which are commonly employed in CF RecSys research. Based on the results of this evaluation, the inclusion of the presented step to the algorithm presented in [7] has proven to significantly upgrade recommendation quality, while maintaining high coverage levels.

The remainder of the article is structured as follows. In Section 2, contemporary related research is summarized. In Section 3, we outline the concept of RPCFs, for self-containment of the article. In Section 4, we present the proposed approach. In Section 5, we display and analyze its evaluation output. Finally, in Section 6, we conclude the article and outline future work.

2 Related Work

CF recommendation quality is a subject that numerous works have researched over the last 10 years.

The work in [8] presents a new similarity metric. The mathematical formulation is developed using two main parts. The first is converting intuitive and qualitative conditions that the similarity metric must meet into appropriate equations, including a combination of integral and differential formulas. and the second is solving these formulas to derive the final similarity metric function.

The work in [9] presents a multi-level recommendation strategy designed to assist users in making decisions by providing more accurate recommendations. This approach can be applied to various online platforms that use CF RecSys, improving the overall user experience. Its effectiveness is shown through an extensive experimental evaluation using real-world datasets.

The study in [10] presents a weighted clustering CF RecSys. This system is built to provide effective recommendations with low cost of computations. Unlike traditional incremental methods, the complexity of this approach is not determined by the number of items and users in the rating database. As a result, this CF RecSys is ideal for dynamic environments with large databases, where information changes quickly, such as introduction of new ratings, updating of existing ratings and input of new items and users.

The work in [11] presents Relational CF to utilize the connections between multiple items in RecSys. It highlights the importance of both the relation type and value predicting preferences of users. In order to tackle this issue, this work proposes a two-level hierarchical attention algorithm for modeling user preferences. The first level emphasizes the significance of relation types, while the second level concentrates on specific relation values to evaluate the contributions of items.

The work in [12] introduces a hybrid CF method that integrates semantic with social recommendations. This method incorporates two classification policies to enhance recommendation success. The proposed method applies the incremental K-means technique to the users and then uses the KNN method to handle new users.

The work in [13] presents a hybrid RecSys algorithm that combines CF and content-based filtering. This algorithm takes advantage of the strengths of content filtering while enabling CF for all items, even those that have not been rated. These unrated items can still be filtered and recommended to users, thus addressing the issue of CF cold start. As the number of users and rating increases, the CF rating matrix becomes denser, which reduces sparsity and enhances the accuracy of CF rating predictions. By combining both methods, the CF RecSys performance is improved.

The study in [14] introduces a new CF algorithm that leverages user preference clustering to address the issue of data sparsity. First, user groups are formed to distinguish between users with different preferences. Then, based on the preferences of the AU, the NNs are determined from the relevant user groups. Furthermore, this work proposes a new similarity metric to compute user proximity more accurately, by considering user preferences from both local and global level.

The work in [15] proposes DeepICF, an item-based CF method that accounts for higher-order and nonlinear relations among products. Instead of only considering the similarity between two products, the method incorporates interactions between all product pairs utilizing nonlinear neural networks. This approach enables the method to successfully capture the higher-order relationships between products, uncovering more complex influences on user decision-making.

The work in [16] proposes an algorithm that combines deep interaction modeling and deep learning techniques for RecSys that rely on a rating matrix, namely Hybrid Neural CF algorithm. This approach aggregates ratings from multiple external sources to address the issue of CF cold start. It achieves this by employing a multivariate rating system that aggregated star ratings and sentiment scores. To overcome the identified challenges, the proposed algorithm consists of four main modules, the Hierarchical User and Product and User Attention, the Deep Multivariate Rating, the Deep CF, and the Neural Sentiment Classifier.

The work in [17] presents a deep neural network method that improves CF prediction results. It utilizes the concept of reliability to enhance the quality of predictions by integrating the prediction errors into the deep learning layers. The core idea of this method is to recommend products that both have high rating prediction values and are considered reliable at the same time. The proposed method is organized into three stages: the reliability prediction, the error prediction, and the ratings prediction. Each of these involves a learning procedure and offer a different abstraction level.

The work in [18] proposes a deep learning CF algorithm, namely CbDNCF. Initially, CbDNCF filters the dataset through a preprocessing layer, followed by training the cleaned data using a classification layer. Next, it performs feature extraction and predicts the highest ratings. The incorporation of Chimp functions into the deep neural classification layer allows the model to achieve optimal rating prediction results.

The work in [19] introduces LDCF, a neural location-aware CF technique designed for recommending web services. This technique maps service location features to high-

dimensional embedding vectors. Additionally, it incorporates a multilayer neural network to capture nonlinear and high-dimensional attributes, along with a similarity corrector to adjust the predicted QoS characteristics. Therefore, the proposed technique effectively addresses the problem of data sparsity and learns the high-dimensional relationships between web services and users.

The work in [20] proposes a deep learning-based method that integrates CF with textual content in order to produce web services recommendations. The interaction between the functionalities of mashups and services is integrated into a neural network, allowing the method to capture the relationships between mashups and services, using a sparse matrix of interactions.

Still, the inclusion of rating prediction confidence factors in the CF recommendation formulation procedure has not received considerable research attention.

Towards this direction, the work in [21] proposes a CF algorithm that declines products bearing low RPCFs values from becoming recommendations to users. Despite the high recommendation quality enhancement this algorithm accomplished, it suffers from significant coverage reduction, making this algorithm inapplicable to sparse datasets (such as the Ciao dataset, the Epinions dataset and Amazon-sourced ones, used in this work). Since the work in [21] strictly enforces the requirement that a prediction should fulfil all RPCFs to qualify for the recommendation formulation stage, in the rest of this paper, we will refer to the algorithm presented in [21] as Strict RPCF Filtering (S-RCPF-F).

The work in [7] introduces a CF recommendation algorithm that is comprised of 4 steps. In the first step, the RPCFs number satisfied by each rating prediction in the ICRL is calculated. In the second step, the set of products is partitioned into 3 subsets, based on the assumption that the user will “definitely” (subset namely “Top”), “most probably” (subset namely “Med”) and “probably” (subset namely “Low”) like. The criterion for this partition is solely the CF prediction values of the products ([4.5-5.0], [4.0-4.5), and [3.5-4.0), for the “Top”, “Med” and “Low” subsets, respectively). In the third step, the products within each subset are sorted in descending order of their number of the RPCFs number satisfied. Lastly, in the fourth step, the products are recommended to the users, starting from the “Top” dataset, based on the sorting performed in the third step (in case the products in the “Top” subset do not suffice, the products of the “Med”, primarily, and “Low”, secondarily, subsets are recommended, using the same sorting). Although this algorithm accomplishes zero recommendation coverage reduction, it achieves mediocre recommendation quality upgrade (2-3% both in terms of recommendation precision and real rating value).

The present work introduces an extension to the algorithm presented in [7] that leverages RPCFs to further enhance recommendation quality. The proposed extension removes from the ICRL the items whose predictions do not fulfil any confidence factor, despite their rating prediction values, and subsequently proceeds in formulating the recommendation with the remaining elements of the ICRL, exactly as the algorithm presented in [7] performs (i.e., the second, third, and fourth step of the algorithm presented in [7]). Under this approach, only predictions that have increased accuracy guarantees are utilized, and therefore recommendation quality is enhanced. Contrary to the algorithm presented in [21], it achieves low recommendation coverage reduction, while at

the same time it can be applied to every CF dataset, since it is based only on the essential collaborative filtering data (i.e., rating records which include the user id, the item id, and the respective rating).

3 Rating Prediction Confidence Factors in CF

In this Section, the prerequisites of this work, concerning the RPCFs in CF are summarized, for self-containment.

Recent studies have explored the connection between basic rating prediction features and the accuracy of rating predictions in the CF process. It is established that the accuracy of rating predictions can be estimated in advance, with a high degree of certainty. Specifically, the studies in [5,6] asserted a direct relationship between CF rating prediction accuracy and three key factors (i.e., rating prediction features): (i) the NN number involved in the rating prediction calculation, denoted as NN_{factor} , (ii) the mean value of the AU's ratings, denoted as $AVGU_{factor}$, and (iii) the mean value of the item's ratings, denoted as $AVGI_{factor}$.

In regard to the dense datasets (i.e., density $> 1\%$), it has been found that a rating prediction in CF is regarded as “highly accurate” if:

- $NN_{factor} \geq 15\%$ (i.e., the percentage of the user's NNs that take part in the prediction formulation is $\geq 15\%$),
- $AVGU_{factor} \geq 4.5$ OR $AVGU_{factor} \leq 1.5$ (when using a 5-star rating scale, as happens in the majority of the RecSys datasets), and
- $AVGI_{factor} \geq 4.5$ OR $AVGI_{factor} \leq 1.5$ (again, when using a 5-star rating scale).

On the other hand, as far as sparse datasets are concerned (i.e., datasets with density $< 1\%$) it has been found that a CF rating prediction is considered as “highly accurate” if:

- $NN_{factor} \geq 4$ (i.e., at least 4 of the user's NNs take part in the prediction formulation),
- $AVGU_{factor} \geq 4.5$ OR $AVGU_{factor} \leq 1.5$ (when using a 5-star rating scale), and
- $AVGI_{factor} \geq 4.5$ OR $AVGI_{factor} \leq 1.5$ (again, when using a 5-star rating scale).

Notably, all rules involving rating average thresholds can be generalized for use with any rating scale, by setting the inclusion conditions to comprise users/items whose mean rating value falls within the higher 10% of the rating scale or the lower 10% of the rating scale.

Having this information at hand, in the next Section we present the extension to the algorithm presented in [7] that leverages RPCFs to further enhance recommendation quality. The algorithm in [7] will be denoted as RPCF-S, since it sorts recommendation according to their RCPFs, without performing any filtering.

4 The Proposed Extension

As noted above, the present work introduces an extension to the algorithm presented in [7], that leverages RPCFs to further enhance recommendation quality. The proposed extension removes from the ICRL the items whose predictions do not fulfil any confidence factor, despite their rating prediction values, and subsequently proceeds to formulate the recommendation with the remaining elements of the ICRL.

Practically, the proposed extension inserts a procedure between the first and second step of the RPCF-S algorithm. This procedure considers the RPCFs that each rating prediction fulfills (calculated in the first step of the RPCF-S algorithm) and eliminates predictions that do not fulfill any RPCFs. On the contrary, the rating predictions that fulfill at least one of the RPCFs are forwarded to the second step of the RPCF-S algorithm (i.e., the partitioning into the 3 subsets “Top”, “Med”, and “Low”, and so on), in order to proceed with the recommendation process.

Under this approach, only predictions that have (relatively) increased accuracy guarantees are utilized, and therefore the quality of the recommendations is enhanced. A graphical representation of the proposed algorithm is depicted in Fig. 1.

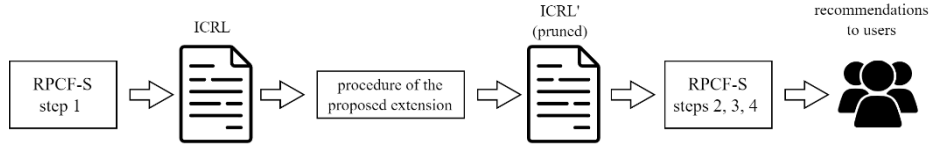


Fig. 1. Activity diagram for the proposed algorithm operation, including the original steps of the RPCF-S algorithm and the proposed extension.

The RPCF-S algorithm, including the proposed extension, achieves upgraded recommendation quality, and maintains high recommendation coverage. Furthermore, this combination can be applied in every CF dataset, since it is based only on the essential CF data (i.e., rating records which include the user id, the item id, and the respective rating).

In the next Section, we experimentally assess the recommendation accuracy of the presented algorithm.

5 Experimental Evaluation

In this section, after presenting the datasets used in the evaluation and the experimental settings (subsection 5.1), we detail the results of the experiments. More specifically, in subsection 5.2, we report on the results concerning recommendation coverage, demonstrating the viability of the proposed extension. Subsequently, in subsection 5.3, we present the experimental outcomes regarding recommendation quality enhancement, substantiating the effectiveness of the algorithm. Lastly, the execution overhead of the presented extension is analyzed in subsection 5.4, to confirm its applicability in terms of execution efficiency.

5.1 Experimental Settings

In our experimental evaluation we use eight CF datasets, including sparse, dense, and social datasets, to cover all sparsity levels. These datasets are broadly used in CF research and their characteristics are outlined in Table 1.

Table 1. The characteristics of the datasets used in the experimental evaluation.

Dataset Name	Dataset Type	Dataset Attributes (#ratings, #users, #items, density, rating range)	CF Category
Amazon Videogames [22]	e-commerce	473K, 17.5K, 55K, 0.05%, 1-5	Sparse
Amazon Digital Music [22]	e-commerce	145K, 12K, 17K, 0.07%, 1-5	Sparse
Amazon Musical Instruments [22]	e-commerce	231K, 28K, 11K, 0.08%, 1-5	Sparse
Amazon CDs and Vinyl [22]	e-commerce	1.44M, 112K, 74K, 0.02%, 1-5	Sparse
MovieLens 100K [23]	media consumption	100K, 600, 9.7K, 1.72%, 0.5-5	Dense
MovieLens 1M [23]	media consumption	1M, 6K, 3.7K, 4.5%, 1-5	Dense
CiaoDVD [24]	social-trust network	73K, 17.6K, 16K, 0.03%, 1-5	Sparse, Social
Epinions [25]	social-trust network	665K, 49K, 134K, 0.01%, 1-5	Sparse, Social

Regarding the user similarity metrics, in this work we use the two most commonly used ones in CF research, i.e., the Pearson Correlation (PC) and the Cosine Similarity (COS) metrics [26–29].

Moreover, for the NN selection methods, we use the top-K NN (KNN) and the threshold method (THR). Following the approaches of the works in [7,21,30,31], for the KNN method, we have set $K=200$ and $K=500$ in our experiments. For the THR method we have set $T=0.0$ and $T=0.5$.

Afterwards, a 5-fold cross validation process [7,21] commences, for predicting rating values on the datasets outlined in Table 1, in which the extended algorithm was set to generate 3 and 5 recommendations per user (top-3 and top-5).

The quality of these recommendations was assessed using the following evaluation metrics:

1. the recommendation precision,
2. the mean real arithmetic rating value, and
3. the NDCG (Normalized Discounted Cumulative Gain)

following the works in [7,21].

Lastly, in order to compute the recommendation precision and NDCG quality metrics, the items having real rating values $\geq 3.5/5$ for the AU are considered as liked/approved by him/her [5,7,32–34].

5.2 Evaluating the applicability of the proposed extension

In this subsection we report on our experimental findings for the recommendation coverage achieved by the proposed algorithm, an aspect that is critical to its applicability, since a recommender system must offer personalized recommendations to the widest possible set of its users.

Fig. 2 illustrates the recommendation coverage, when recommending 3 items to each user (top-3 recommendations) and utilizing the PC similarity metric, with the KNN selection method and setting $K=200$. More specifically, Figure 2 illustrates the percentage of users where each algorithm is able to generate a complete recommendation (i.e., recommending 3 items) for each user in the dataset. The results obtained for the proposed algorithm are compared against the performance of the RCPF-S algorithm (without the proposed extension), since it achieves no coverage reduction in relation to the plain CF algorithm, and the performance of the S-RCPF-F algorithm [21], which also undertakes a confidence factor-based approach, however performing more aggressive filtering on candidate recommendation items.

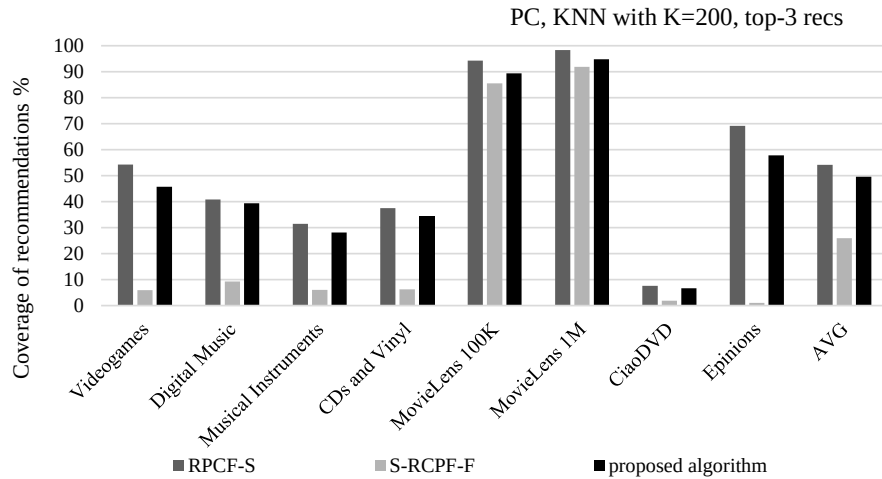


Fig. 2. Recommendation coverage considering the top-3 recommendations, using the PC similarity metric and the KNN selection method.

We can observe that, contrary to the S-RCPF-F algorithm [21], the RCPF-S algorithm, including the extension proposed in this work, maintains high recommendation coverage. More specifically, while the average recommendation coverage of the algorithm presented in [21] is 26% (corresponding to a reduction equal to 52%, in comparison to the RCPF-S algorithm), the average recommendation coverage of the proposed algorithm is 49.6%, thus sustaining a significantly lower reduction of 8.5%. Regarding the sparse datasets, where the S-RCPF-F algorithm [21] is proven inapplicable, the coverage reduction of the proposed algorithm is less than 17% in each of the datasets tested. A notable case is the Epinions dataset, in which the application of the S-RCPF-F

algorithm [21] leads a recommendation coverage decline from 69% to 1%), while the proposed algorithm maintains coverage at the level of 58%.

As depicted in Fig. 3, similar results are observed when the COS similarity metric is used, under the same settings (i.e., KNN selection method, $K=200$ and recommending 3 items). More specifically, the average coverage reduction is measured at 11.5%, while the respective coverage reduction of the S-RCPF-F [21] is 65%.

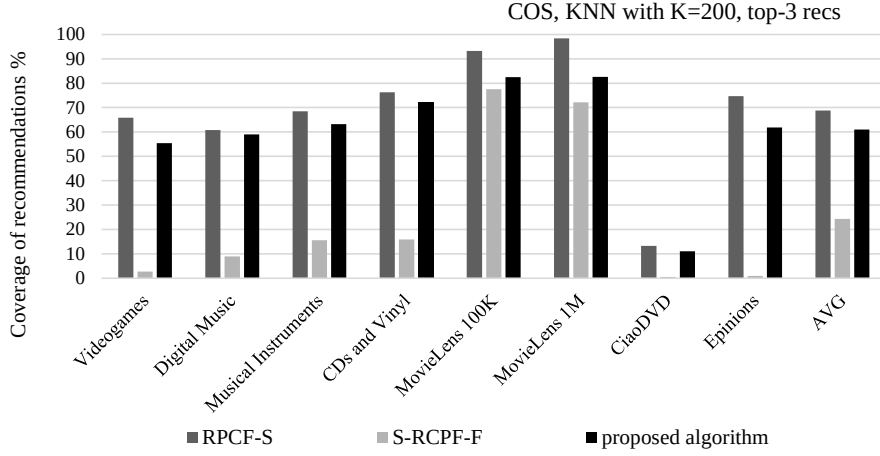


Fig. 3. Recommendation coverage considering the top-3 recommendations, using the COS similarity metric and the KNN selection method.

Similar recommendation coverage results are measured when using the PC similarity metric, with the THR NN selection method with $T=0.0$ and recommending 3 items. In this case, the average coverage reduction is measured at 6.8%, while the respective average coverage reduction of the S-RCPF-F algorithm [21] is 51%. Similarly, when the COS similarity metric is used, under the exact same settings, the average coverage reduction is measured at 5%, while the respective coverage reduction of the S-RCPF-F algorithm [21] is 55%.

In our experimental evaluation we have used all the combinations of parameters listed in subsection 5.1 (i.e., user similarity computation metric, NN selection method with varying hyperparameters, number of recommended items). All results are in close agreement with the ones reported above and their detailed presentation is omitted for conciseness.

The recommendation coverage results detailed above indicate the applicability of the proposed extension, both in dense and in sparse datasets. Furthermore, considering that the S-RCPF-F algorithm [21] is demonstrated to be inapplicable on sparse datasets, due to excessive recommendation coverage reduction, it will not be considered as a comparison baseline in the experiment results presented in subsection 5.3, which aim at assessing the recommendation quality enhancement of the proposed approach. In place of the S-RCPF-F algorithm [21], the plain CF algorithm will be used as a baseline.

5.3 Evaluating the recommendation quality enhancement of the proposed extension

After having established the applicability of the proposed extension, both in dense and in sparse datasets, we proceed in evaluating its performance in terms of recommendation accuracy/success. More specifically, we compare the performance of the proposed algorithm both with the plain CF algorithm and the original RPCF-S algorithm (without the inclusion of this extension).

Fig. 4 illustrates the average recommendation precision, when recommending 3 items to each user and utilizing the PC similarity metric with the KNN selection method and setting $K=200$. The proposed algorithm increases the recommendation precision by 6.7%, in comparison to the plain CF algorithm, achieving a substantial improvement compared to the respective increase of the original RPCF-S algorithm, which is measured at 2.5%.

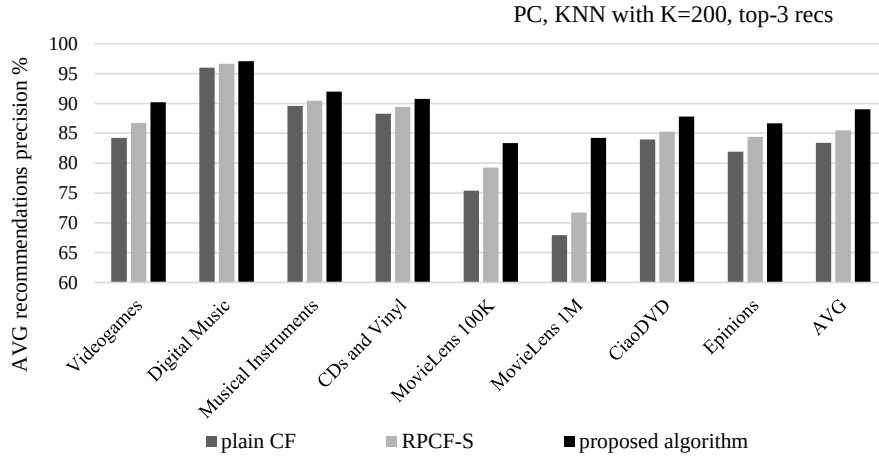


Fig. 4. Mean precision considering the top-3 recommendations, utilizing the PC similarity metric and the KNN selection method.

Fig. 5 illustrates the respective measurements when the COS similarity metric is used, while keeping all the other settings identical. The results in Fig. 5 are very similar to those in Fig. 4. More specifically, the proposed algorithm increases the recommendation precision by 7.4%, in comparison to the plain CF algorithm. These gains are considerably higher than the ones achieved by the original RPCF-S algorithm, which are quantified at 2.3%.

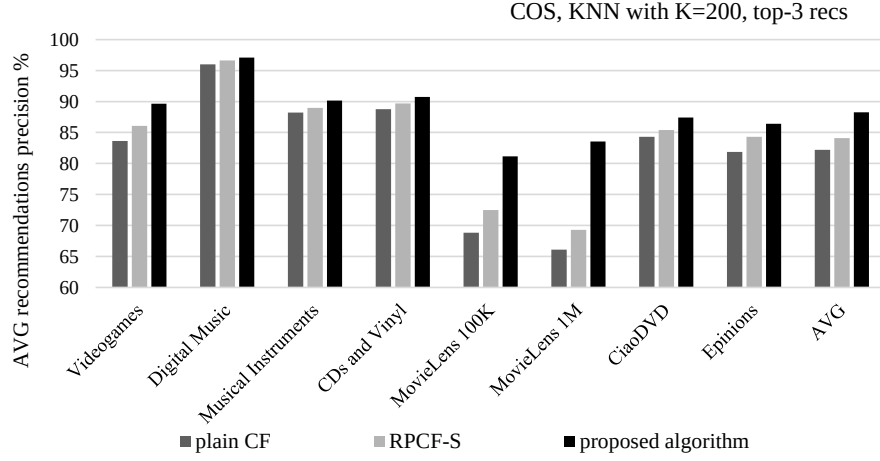


Fig. 5. Mean precision considering the top-3 recommendations, utilizing the COS similarity metric and the KNN selection method.

Similar results are observed when using the PC similarity metric, with the THR NN selection method with $T=0.0$ and recommending 3 items. In this case, the average precision enhancement compared to the plain CF algorithm is measured at 3.1%, while the respective increase of the original RPCF-S algorithm is measured at 1.3%. Correspondingly, when the COS similarity metric is used under the exact same settings, the average precision enhancement achieved by the proposed algorithm is measured at 2.5%, while the respective increase of the original RPCF-S algorithm which is measured at 1.3%.

Fig. 6 depicts the mean real rating value, when recommending 3 items to each user and utilizing the PC similarity metric and the KNN selection method, setting $K=200$. The proposed algorithm increases the average real rating value by 4% in comparison to the plain CF algorithm, surpassing the respective increase of the original RPCF-S algorithm, which is measured at 1.6%.

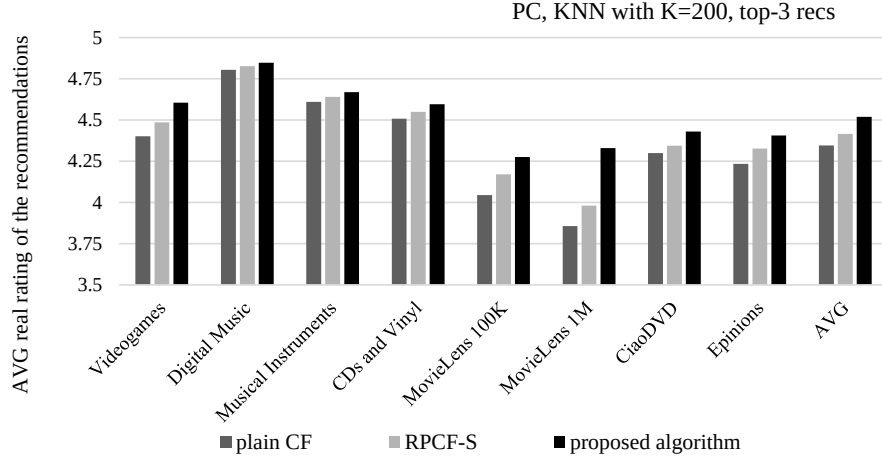


Fig. 6. Mean real rating value considering the top-3 recommendations, utilizing the PC similarity metric and the KNN selection method.

Fig. 7 illustrates the respective findings when the COS similarity metric is used, while all other settings remain unaltered. The results in Fig. 7 are highly similar to those presented in Fig. 6, and more specifically, the proposed algorithm increases the average real rating value by 4.4% in comparison to the plain CF algorithm, surpassing the respective increase of the original RPCF-S algorithm, which is measured at 1.5%.

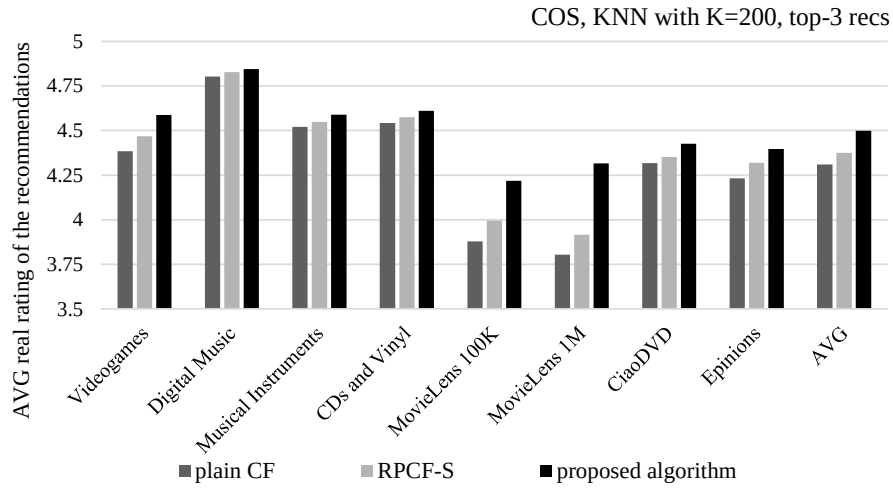


Fig. 7. Mean real rating value considering the top-3 recommendations, utilizing the COS similarity metric and the KNN selection method.

Analogous real rating value results are measured when using the PC similarity metric, with the THR NN selection method with $T=0.0$ and recommending 3 items. In this

case, the proposed algorithm improves the mean real rating value of recommendations from 4.45/5 to 4.55/5, surpassing the respective value achieved by original RPCF-S, measured at 4.49. Similarly, when the COS similarity metric is used under the exact same settings, the proposed algorithm augments the mean real rating value of recommended items from 4.43 to 4.51/5, while the original RPCF-S algorithm attains an increment up to 4.48/5.

Fig. 8 illustrates the average NDCG value of recommendations, when recommending 3 items to each user and utilizing the PC similarity metric with the KNN selection method, setting $K=200$. Considering the average of all eight datasets, the proposed algorithm improves the mean NDCG value of recommendations from 0.973 to 0.983, while the RPCF-S algorithm attains an increment up to 0.978.

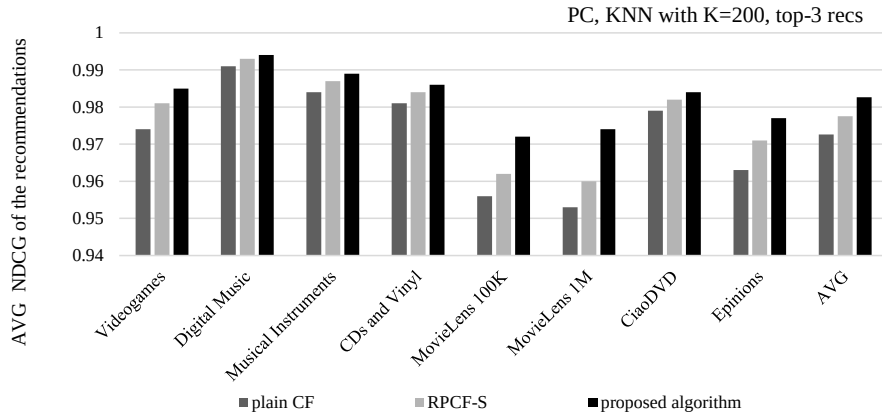


Fig. 8. Mean NDCG considering the top-3 recommendations, utilizing the PC similarity metric and the KNN selection method.

Fig. 9 illustrates the respective output when the COS similarity metric is used, while all the settings remain the same. The results in Fig. 9 are in agreement to those of Fig. 8. More specifically, the proposed algorithm is able to increase the mean NDCG value of recommendations from 0.969 to 0.981, surpassing the respective value achieved by the original RPCF-S algorithm, which is measured at 0.974.

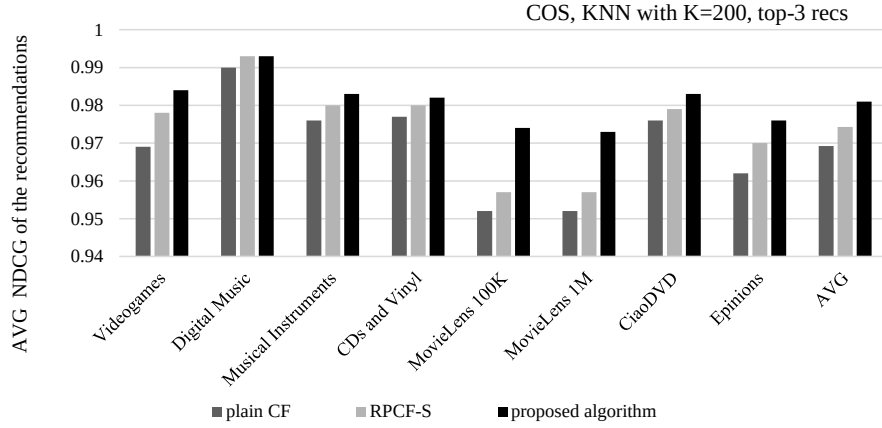


Fig. 9. Mean NDCG considering the top-3 recommendations, utilizing the COS similarity metric and the KNN selection method.

Similar NDCG value results are measured when using the PC similarity metric, with the THR NN selection method with $T=0.0$ and recommending 3 items. In this case, the proposed algorithm is able to increase the mean NDCG value of recommendations from 0.976 to 0.982, exceeding the respective value achieved by RPCF-S algorithm, which is measured at 0.98. Similarly, when the COS similarity metric is used under the exact same settings, the mean NDCG value is enhanced from 0.974 to 0.980, while the respective value of the original RPCF-S algorithm is measured at 0.978.

All the combinations of parameters listed in subsection 5.1 (i.e., user similarity computation metric, NN selection method with varying hyperparameter values, number of recommended items) have been experimentally tested, and the results obtained are in close agreement with the ones reported above. Their detailed presentation is omitted for conciseness.

The findings reported in this subsection substantiate that the proposed algorithm achieves considerable recommendation quality enhancement, both in dense and in sparse datasets.

5.4 Execution Overhead of the Presented Extension

The proposed extension introduces an additional overhead, when compared to the original RPCF-S algorithm presented in [7], due to the inclusion of the procedure which rejects from the ICRL the rating predictions that do not fulfill any RPCFs (c.f. Fig. 1, Section 4). However, the overhead incurred due to the introduction of the additional step is negligible, since this step consists of a simple comparison for each rating prediction. Notably, since the ICRL is pruned to remove predictions that do not fulfill any RPCFs, steps 2 and 3 of the RPCF-S algorithm (partitioning the pruned ICRL to “Top”, “Med” and “Low” classes, and sorting predictions in each subset, respectively) can be performed in shorter time (due to the decreased length of the ICRL).

The execution time of the proposed algorithm has been measured for all eight datasets used in this work. The obtained measurements indicate that the extended algorithm proposed in this paper exhibits execution time gains from 2% to 15% in comparison to the RPCF-S algorithm (depending on the dataset and the execution settings and effectively mapping to the percentage of ICRL elements which are eliminated). This finding corroborates the applicability of the proposed extension.

6 Conclusion and Future Work

This work proposes an extension to the RPCF-S algorithm presented in [7] that leverages RPCFs to further enhance recommendation quality. More specifically, the proposed extension removes from the ICRL the items whose predictions do not fulfil any confidence factor, and subsequently proceeds to formulate the recommendation with the remaining elements of the ICRL. Under this approach, only predictions that have increased accuracy guarantees are utilized, and therefore the quality of the recommendation is enhanced.

A multi-parameter set of experiments is used to thoroughly assess the proposed extension for both applicability and recommendation quality. Regarding its applicability, the inclusion of the proposed extension (i) maintains high rating recommendation coverage, (ii) reduces the recommendation formulation time, and (iii) it can be applied in every CF dataset, since it is based only on the essential CF data.

Furthermore, the inclusion of the proposed extension has been shown to significantly upgrade the recommendation quality in all datasets and settings (i.e., similarity metrics, NN selection methods and number of recommendations produced per user), based on all three qualitative metrics used, namely (i) recommendation precision, (ii) the real rating of the recommendations, and (iii) NDCG values.

Regarding our future work, we will focus on investigating more CF prediction accuracy-related features and apply them into the recommendation procedure. Furthermore, we are planning to incorporate additional RecSys information sources (e.g., users' social information and demographics, attributes and categories of items, etc.) into the CF rating prediction and recommendation processes.

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References

1. Wu, L., He, X., Wang, X., Zhang, K. & Wang, M. A Survey on Accuracy-oriented Neural Recommendation: From Collaborative Filtering to Information-rich Recommendation. *IEEE Trans. Knowl. Data Eng.* 1–1 (2022) doi:10.1109/TKDE.2022.3145690.

2. Koren, Y., Rendle, S. & Bell, R. Advances in Collaborative Filtering. in *Recommender Systems Handbook* (eds. Ricci, F., Rokach, L. & Shapira, B.) 91–142 (Springer US, New York, NY, 2022). doi:10.1007/978-1-0716-2197-4_3.
3. Xie, Z. *et al.* Neighborhood-Based Collaborative Filtering for Conversational Recommendation. in *18th ACM Conference on Recommender Systems* 1045–1050 (ACM, Bari Italy, 2024). doi:10.1145/3640457.3688191.
4. Papadakis, H., Papagrigoriou, A., Panagiotakis, C., Kosmas, E. & Fragopoulou, P. Collaborative filtering recommender systems taxonomy. *Knowl Inf Syst* **64**, 35–74 (2022).
5. Margaritis, D., Vassilakis, C. & Spiliotopoulos, D. On Producing Accurate Rating Predictions in Sparse Collaborative Filtering Datasets. *Information* **13**, 302 (2022).
6. Spiliotopoulos, D., Margaritis, D. & Vassilakis, C. On Exploiting Rating Prediction Accuracy Features in Dense Collaborative Filtering Datasets. *Information* **13**, 428 (2022).
7. Sgardelis, K., Margaritis, D., Spiliotopoulos, D., Vassilakis, C. & Ougiaroglou, S. Improving Recommendation Quality in Collaborative Filtering by Including Prediction Confidence Factors: in *Proceedings of the 20th International Conference on Web Information Systems and Technologies* 372–379 (SCITEPRESS - Science and Technology Publications, Porto, Portugal, 2024). doi:10.5220/0013052200003825.
8. Gazdar, A. & Hidri, L. A new similarity measure for collaborative filtering based recommender systems. *Knowledge-Based Systems* **188**, 105058 (2020).
9. Polatidis, N. & Georgiadis, C. K. A multi-level collaborative filtering method that improves recommendations. *Expert Systems with Applications* **48**, 100–110 (2016).
10. Salah, A., Rogovschi, N. & Nadif, M. A dynamic collaborative filtering system via a weighted clustering approach. *Neurocomputing* **175**, 206–215 (2016).
11. Xin, X., He, X., Zhang, Y., Zhang, Y. & Jose, J. Relational Collaborative Filtering: Modeling Multiple Item Relations for Recommendation. in *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* 125–134 (ACM, Paris France, 2019). doi:10.1145/3331184.3331188.
12. Ramakrishna, M. T. *et al.* HCoF: Hybrid Collaborative Filtering Using Social and Semantic Suggestions for Friend Recommendation. *Electronics* **12**, 1365 (2023).
13. Li, L., Zhang, Z. & Zhang, S. Hybrid Algorithm Based on Content and Collaborative Filtering in Recommendation System Optimization and Simulation. *Scientific Programming* **2021**, 1–11 (2021).
14. Zhang, J., Lin, Y., Lin, M. & Liu, J. An effective collaborative filtering algorithm based on user preference clustering. *Appl Intell* **45**, 230–240 (2016).
15. Xue, F. *et al.* Deep Item-based Collaborative Filtering for Top-N Recommendation. *ACM Trans. Inf. Syst.* **37**, 1–25 (2019).
16. Ibrahim, M., Bajwa, I. S., Sarwar, N., Hajjej, F. & Sakr, H. A. An Intelligent Hybrid Neural Collaborative Filtering Approach for True Recommendations. *IEEE Access* **11**, 64831–64849 (2023).

17. Bobadilla, J., Alonso, S. & Hernando, A. Deep Learning Architecture for Collaborative Filtering Recommender Systems. *Applied Sciences* **10**, 2441 (2020).
18. Mandalapu, S. R., Narayanan, B. & Putheti, S. A hybrid collaborative filtering mechanism for product recommendation system. *Multimed Tools Appl* **83**, 12775–12798 (2023).
19. Zhang, Y., Yin, C., Wu, Q., He, Q. & Zhu, H. Location-Aware Deep Collaborative Filtering for Service Recommendation. *IEEE Trans. Syst. Man Cybern., Syst.* **51**, 3796–3807 (2021).
20. Xiong, R., Wang, J., Zhang, N. & Ma, Y. Deep hybrid collaborative filtering for Web service recommendation. *Expert Systems with Applications* **110**, 191–205 (2018).
21. Margaris, D., Sgardelis, K., Spiliotopoulos, D. & Vassilakis, C. Exploiting Rating Prediction Certainty for Recommendation Formulation in Collaborative Filtering. *BDCC* **8**, 53 (2024).
22. Ni, J., Li, J. & McAuley, J. Justifying Recommendations using Distantly-Labeled Reviews and Fine-Grained Aspects. in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* 188–197 (Association for Computational Linguistics, Hong Kong, China, 2019). doi:10.18653/v1/D19-1018.
23. Harper, F. M. & Konstan, J. A. The MovieLens Datasets: History and Context. *ACM Trans. Interact. Intell. Syst.* **5**, 1–19 (2016).
24. Guo, G., Zhang, J., Thalmann, D. & Yorke-Smith, N. ETAF: An extended trust antecedents framework for trust prediction. in *2014 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2014)* 540–547 (IEEE, China, 2014). doi:10.1109/ASONAM.2014.6921639.
25. Ma, N., Lim, E.-P., Nguyen, V.-A., Sun, A. & Liu, H. Trust relationship prediction using online product review data. in *Proceedings of the 1st ACM international workshop on Complex networks meet information & knowledge management* 47–54 (ACM, Hong Kong China, 2009). doi:10.1145/1651274.1651284.
26. Jain, G., Mahara, T. & Tripathi, K. N. A Survey of Similarity Measures for Collaborative Filtering-Based Recommender System. in *Soft Computing: Theories and Applications* (eds. Pant, M., Sharma, T. K., Verma, O. P., Singla, R. & Sikander, A.) vol. 1053 343–352 (Springer Singapore, Singapore, 2020).
27. Hassanieh, L. A., Jaoudeh, C. A., Abdo, J. B. & Demerjian, J. Similarity measures for collaborative filtering recommender systems. in *2018 IEEE Middle East and North Africa Communications Conference (MENACOMM)* 1–5 (IEEE, Jounieh, 2018). doi:10.1109/MENACOMM.2018.8371003.
28. Chowdhury, P. & Sinha, B. B. Evaluating the Effectiveness of Collaborative Filtering Similarity Measures: A Comprehensive Review. *Procedia Computer Science* **235**, 2641–2650 (2024).
29. Hartatik, Isnanto, R. R., Warsito, B., Firdaus, N. & A’La, F. Y. Towards the Application of Artificial Intelligence: Cosine Similarity in Recommendation Systems Based on Collaborative Filtering. in *2024 7th International Conference of Computer and Informatics Engineering (IC2IE)* 1–5 (IEEE, Bali, Indonesia, 2024). doi:10.1109/IC2IE63342.2024.10747946.

30. Fkih, F. Similarity measures for Collaborative Filtering-based Recommender Systems: Review and experimental comparison. *Journal of King Saud University - Computer and Information Sciences* **34**, 7645–7669 (2022).
31. Wang, D., Yih, Y. & Ventresca, M. Improving neighbor-based collaborative filtering by using a hybrid similarity measurement. *Expert Systems with Applications* **160**, 113651 (2020).
32. Trattner, C., Said, A., Boratto, L. & Felfernig, A. Evaluating Group Recommender Systems. in *Group Recommender Systems* (eds. Felfernig, A., Boratto, L., Stettinger, M. & Tkalčič, M.) 63–75 (Springer Nature Switzerland, Cham, 2024). doi:10.1007/978-3-031-44943-7_3.
33. Margaris, D., Vassilakis, C. & Spiliotopoulos, D. What makes a review a reliable rating in recommender systems? *Information Processing & Management* **57**, 102304 (2020).
34. Felfernig, A., Boratto, L., Stettinger, M. & Tkalčič, M. Evaluating Group Recommender Systems. in *Group Recommender Systems* 59–71 (Springer International Publishing, Cham, 2018). doi:10.1007/978-3-319-75067-5_3.